Pranav Singh

(650) 741-8906 | pranavsingh.us@gmail.com | https://ps1526.github.io | LinkedIn | GitHub | La Jolla, CA, USA

Research Interests: Graph Neural Networks, Geometric Deep Learning, Non-Euclidean Foundation Models, Knowledge Graph Reasoning, Large Language Models

Education

- **University of California San Diego** *Sep 2023 - Jun 2027*
  Bachelor of Science in Data Science La Jolla, CA
  - **Coursework**: Geometry of Data, Reinforcement Learning, Large Language Models, Probabilistic Modeling & ML, Representation Learning, Recommender Systems, Theoretical Foundations of Data Science, Scalable Analytics, Data Structures, Linear Algebra, Statistical Methods, Data Visualization

Experience

- **University of California San Diego - Wang Lab** *Nov 2025 - Present*
  Research Assistant La Jolla, CA
  - **Reduced** neural hierarchical clustering training time from $O(n^5)$ to $O(n^3)$ via tree-level gradient batching for efficient optimizer updates, reducing training time from **20+ hours** to **1 hour**
  - **Achieved provably optimal hierarchical clustering** using $O(n^2 \log n)$ priority-queue algorithm with integer linear programming(ILP) solver, **enabling deterministic approach** to tree reconstruction at scale
  - **Trained PPO-based policy** to minimize Dasgupta cost in hierarchical clustering, enabling **scalable optimization beyond ILP computational limits**
- **Yale University - Graph and Geometric Learning Lab** *Nov 2025 - Present*
  Research Assistant Remote
  - **Built** hierarchical retrieval pipeline using hyperbolic embeddings, training **3** neural architectures across **1M+** documents **enabling tree-structured document organization**, preserving semantic relationships
  - **Reduced** memory requirements by **50%** through advanced spectral clustering, enabling tree construction to scale from **50K** to **450K+** documents
  - **Developed** automated LLM-guided retrieval system using Google Gemini API across **15** tree configurations, **achieving 0.95 hit rate for document retrieval**
- **University of California San Diego** *May 2025 - Present*
  Ledell Research Scholar La Jolla, CA
  - **Adapted** GNN-LLM architectures to utilize heterogeneous GNNs with type-aware embeddings, achieving **0.44** Hit@1 accuracy on pharmaceutical graph-QA tasks (**+0.32** improvement over baseline)
  - **Developed** PharmQA benchmark with **4000+** subgraphs from **87K**-node biomedical knowledge graph using relationship-aware random walks across **12** entity types, **enabling systematic GNN-LLM evaluation**
  - **Achieved 90%** success rate converting RDF graphs to Text-Attributed Graphs through systematic API-based entity mapping, **enabling GNN-LLM integration** with biomedical knowledge retreival from PubMed database
- **University of California San Diego - Chiba Lab** *Sep 2024 - Present*
  Research Assistant La Jolla, CA
  - **Developed** audio alignment pipeline to detect breathing patterns in children's speech, addressing frequency differences between children **(75-500 Hz)** and adults
  - **Implemented** voice quality feature extraction and fine-tuned speaker diarization neural network on **50+** hours of children's audio, **achieving 55% improved speaker separation** that enabled accurate multi-child stress analysis
- **National Nanotechnology Coordination Office** *May 2024 - Aug 2025*
  Research Intern Remote
  - **Engineered** AI-powered query system for **131,388**-node federal nanomaterial knowledge graph from EPA, NIOSH, and CPSC, reducing complexity and improving accuracy
  - **Scaled** semantic mapping tool using ontology alignment and schema matching across EPA, NIOSH, and CPSC federal databases, **reducing cross-agency integration time by 80%**
  - **Developed** ontology-aware negative sampling approach for GNNs, **preventing spurious predictions** by ensuring only scientifically plausible nanomaterial experimental combinations

Projects

- **ResearchKG: Full-Stack Research Tool and Knowledge Graph Generator** *Feb 2025*
  Tools: Next.js, D3.js, Semantic Scholar API, spaCy NLP, GCP Cloud Run, Gemini API, AWS Elasticsearch [GitHub]
  - Engineered citation network generator using Semantic Scholar API and spaCy NLP to map 1000+ research relationships from single papers, including author influence metrics and citation patterns
  - Built interactive Next.js/D3.js visualization platform with cycle detection algorithms to identify circular citation dependencies and research echo chambers
  - Integrated Gemini API and AWS Elasticsearch to analyze citation networks and predict emerging research opportunities through Graph RAG
- **Joke Recommendation System: Graph Neural Network Recommender System** *Jul 2025*
  Tools: PyTorch Geometric, Sentence-BERT, TF-IDF, LDA, spaCy, Python [GitHub]
  - Built automated data pipeline aggregating 3,000+ jokes from multiple APIs with rate limiting and comprehensive error handling
  - Engineered multi-dimensional graph relationships using Sentence-BERT embeddings, TF-IDF vectorization, LDA topic modeling, and spaCy entity extraction for semantic similarity analysis
  - Implemented PyTorch Geometric HGT model achieving 82.7% AUC on link prediction tasks
- **Yaadein: AI Memory Preservation Platform** *Oct 2024*
  Tools: Next.js, Deepgram, LangChain, Pinecone, Clerk
  - Built backend architecture leveraging Next.js, Deepgram, LangChain, and Pinecone for AI-powered memory preservation
  - Integrated multi-factor authentication and passwordless login using Clerk to safeguard sensitive memory data

Presentations and Publications

C=Conference, J=Journal, P=Presentation, S=In Submission, T=Thesis

[S.1] Singh, P., Eldridge, J. (2026). **PharmQA: A Multi-Hop Pharmacological Reasoning Benchmark for Knowledge Graph Question Answering**. Manuscript submitted to ACL Student Research Workshop.

[S.2] Singh, P.*, Mortensen, H.* et al. (2025). **The Federal NanoEHS Data Landscape: Machine-learning and Large Language Methods to Improve Data Accessibility, Interoperability and Semantic Queries** Manuscript submitted to Nature Data Descriptor Journal.

[C.2] Singh, P., Chiba, A. (2024). **Can AI help predict stress in children's voices?**. Presented at the UCSD Undergraduate Research Conference.

[P.1] Singh, P.*, Mortensen, H.* et al. (2025). **NIH ICCVAM: The Federal NanoEHS Data Landscape**. Presented at the Interagency Coordinating Committee on the Validation of Alternative Methods Public Forum.

Skills

- **Programming Languages:** Python, Java, C++, JavaScript, SQL, SPARQL
- **Machine Learning:** PyTorch, TensorFlow, HuggingFace, LangChain, Scikit-Learn, Keras, CUDA
- **Optimization & Scientific Computing:** Gurobi, SciPy, NetworkX, Spectral Methods
- **Web Development:** Next.js, React.js, Django, Flask, D3.js, HTML, CSS, Tailwind
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Plotly
- **Cloud Technologies:** AWS (S3, Redshift, Glue), Google Cloud Platform, Firebase, Kubernetes

Honors and Awards

- **Virtual Intern Award (VSFS Program)** *2025*
  U.S. Department of State
  - Awarded Virtual Intern Award, 20 interns selected from 1400+ interns, for recognition of contributions to federal nanomaterial research and development of Graph RAG systems for cross-agency data integration and research
- **Ledell Research Scholarship Award** *2025*
  University of California San Diego
  - Scholarship awarded 22 students selected from 1000+ applicants, supporting undergraduate research in science and engineering

Certifications

- **AWS Certified Cloud Practitioner** *Oct 2024*
- **IBM:** Databases and SQL for Data Science with Python *Jan 2024*